

ITAI 1371 Midterm: Stroke Dataset EDA, Preprocessing and SMOTE.

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Repository: <https://github.com/ClayClimate-AI/itai-1371-midterm-stroke-eda>

Dataset

Kaggle stroke prediction data (docs/dataset_url.md), 5110 rows by 12 columns, target stroke.

EDA findings

bmi missing in 138 train rows and not at random, 1 gender Other row, only 13 Never_worked train rows, and stroke at 4.7% positive.

What I did (D1 to D8)

plain random 70/30 split with seed 33; dropped the Other row and id; median fill for bmi plus a bmi_missing flag; full one hot encoding with handle_unknown ignore; StandardScaler on age, glucose and bmi; fill, encode, scale, then SMOTE on train only; plain SMOTE on work_type (k 5, seed 33, float first); repair, range check and an is_synthetic flag.

Balancing

per Prof Rao (Oct 1, 2026), I balanced on work_type. Every group went to 2040 rows (nb04 cell 9), 6624 synthetic rows added.

ML problem

a stroke risk classifier, judged on the untouched test file with recall on stroke cases.

Deliverables

See the README table in the repository.